



FINE JEWELLERY · EST. MMXXVI

OPALÉRA

Named for the opal — the one stone whose fire holds the colour of every gem: the blue of sapphire, the blush of rose gold, the light of diamond and the lustre of pearl. Eighty handcrafted pieces, one house — and a studio to make yours truly yours.



IFM2B10 · GROUP PROJECT · FINAL PRESENTATION · SEPTEMBER 2026

OPALÉRA — The House of Every Gem

A full-stack B2C fine-jewellery store · Quantum Coders · 80 pieces · 92 pages · Node.js backend · manager console

The Team



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Frontend developer



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Backend developer ·
Database creator



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Frontend developer

Dataset: the 80-piece Kaggle jewellery database · necklaces, earrings & pendants · R 959 – R 6 799

Why OPALÉRA Exists

Buying jewellery online is a gamble — fake stones, no recourse, real pieces priced beyond a normal salary. OPALÉRA answers with a maison you can trust.



Certified authentic

Every piece carries its own certificate of authenticity; the maison refunds in full if a stone is ever misrepresented.



Honest prices

Direct-from-atelier pricing from R 959, VAT included, with the comparable boutique estimate and instalment framing on every piece.



The rating system

Star ratings, written patron reviews with a verified-purchase badge, and thumbs votes — visible before you pay.

What We Built

80

PIECES IN THE
DATABASE

92

HTML PAGES

22

API ENDPOINTS

2

USER ROLES

0

NPM DEPENDENCIES



Node.js server

server.js — routes, script password hashing, httpOnly cookie sessions, server-side totals, static hosting of the site



JSON database

data/db.json — users, sessions, products, orders, reviews, wishlists, carts, resets; seeded on first run



Shared JS layer

opalera-data.js — catalogue helpers, the API client, the live catalogue mirror, order tracking; one script for every page



Deployed

GitHub → Render:
opalera.onrender.com serves the same server, site and API

Every Feature Stands Complete



Website design

one gold-on-midnight system, 92 pages



Product catalog

service-fed, search · price bands · sort · tabs



Database

8 collections, 80 seeded products



Web service

22 JSON endpoints, cookie sessions



Product management

add · edit · retire · delete, live on the storefront



Shopping cart

quantities, stock caps, saved per account



Transactions

promo codes, 15% VAT, stock deducted, 3 pay methods



Invoices

tax invoice per order, QR code, printable



User management

register · login · reset · patron & manager roles



Reports

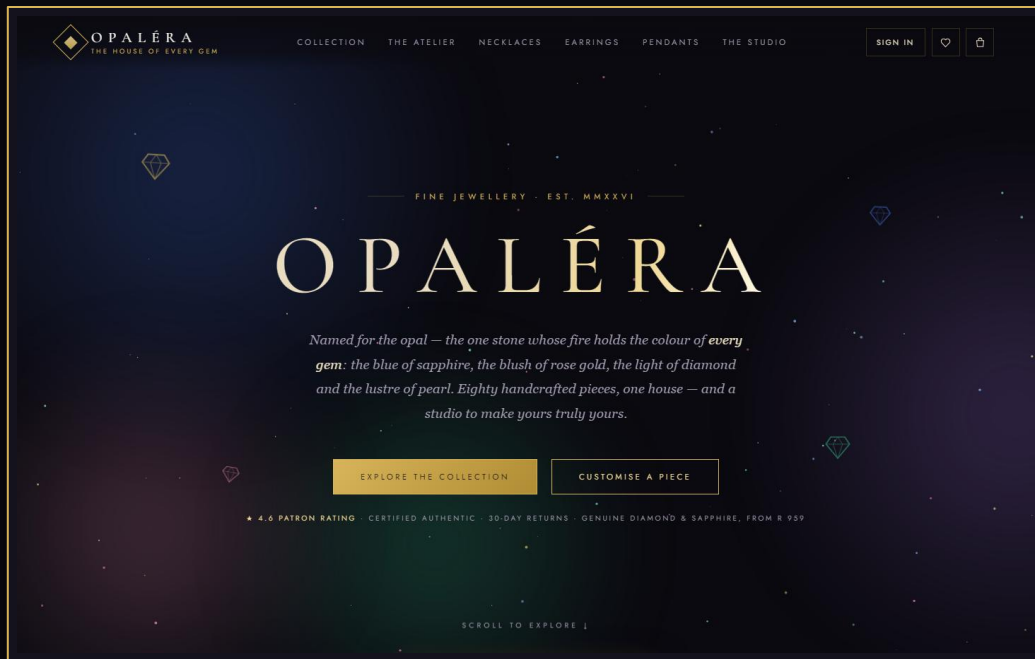
filters, distinct-products control, stock, users/day, 6 KPIs

BONUS · compare pieces side by side · live order-tracking timeline · QR code on every invoice

Arrival at the Maison

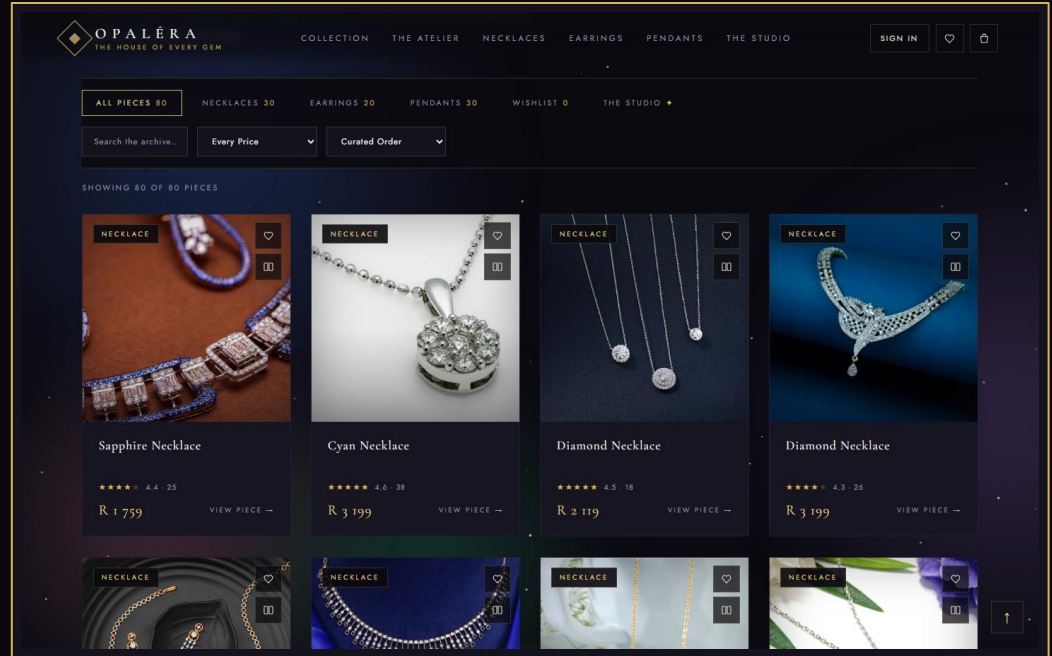
- ◆ Animated aurora, sparkle field and floating gems behind every page
- ◆ One gold-on-midnight design system across all 92 pages
- ◆ Trust strip on arrival: rating · authenticity · returns · from R 959
- ◆ Sign-in, wishlist and bag in the nav — the counts go live the moment a patron signs in

DESIGN · DONE



The Collection — Fed by the Service

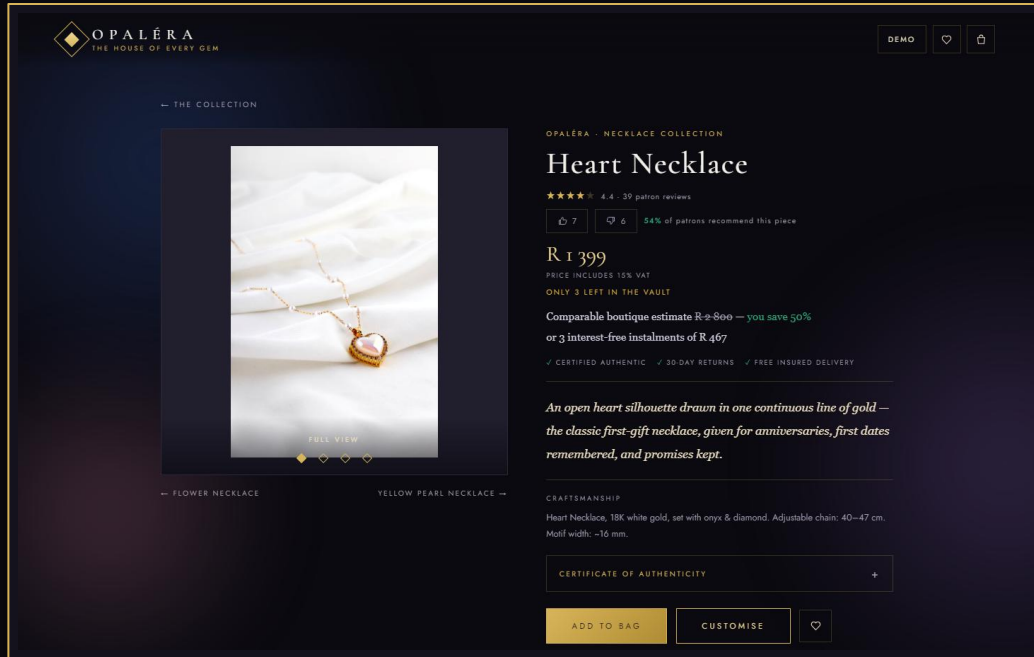
- ◆ The catalogue is retrieved from the database through GET /api/products — prices, photos, stock and retired pieces update live
- ◆ Category tabs with live counts, search, four price bands and sorting by price or name
- ◆ “Showing 80 of 80 pieces” honesty line; wishlist hearts and a compare tick on every card
- ◆ Each card opens the piece's own page



CATALOG · DONE

SERVICE · DONE

The Product Page



- ◆ Hand-written description for all 80 pieces, plus craftsmanship specs
- ◆ Live stock from the database — “Only 3 left in the vault”
- ◆ Stars, thumbs votes and “% recommend”; reviews with the verified-purchase badge
- ◆ Comparable boutique estimate, instalments, certificate of authenticity
- ◆ Manager-added pieces get a page automatically (product.html?id=...)

INFORMATION · DONE

RATINGS · LIVE

Checkout: Quantities, Rules & VAT

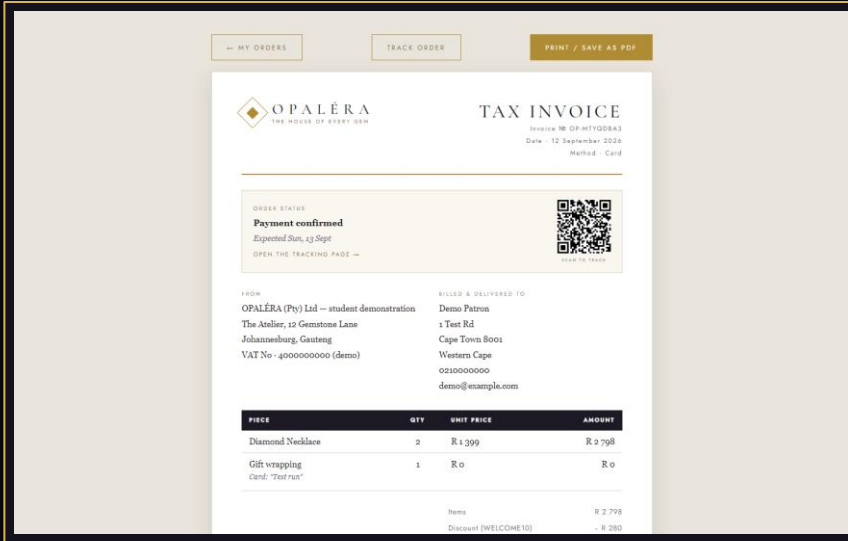
- ◆ Bag with quantities, capped by real stock; customisations itemised
- ◆ Business rules: promo codes (WELCOME10 / GEM10), 15% VAT inside every total, delivery address with all nine provinces
- ◆ Card, Instant EFT or pay-at-counter — totals recomputed on the server so the client can't tamper
- ◆ Stock is deducted on the server the moment the order is placed

CART · DONE

TRANSACTIONS · DONE

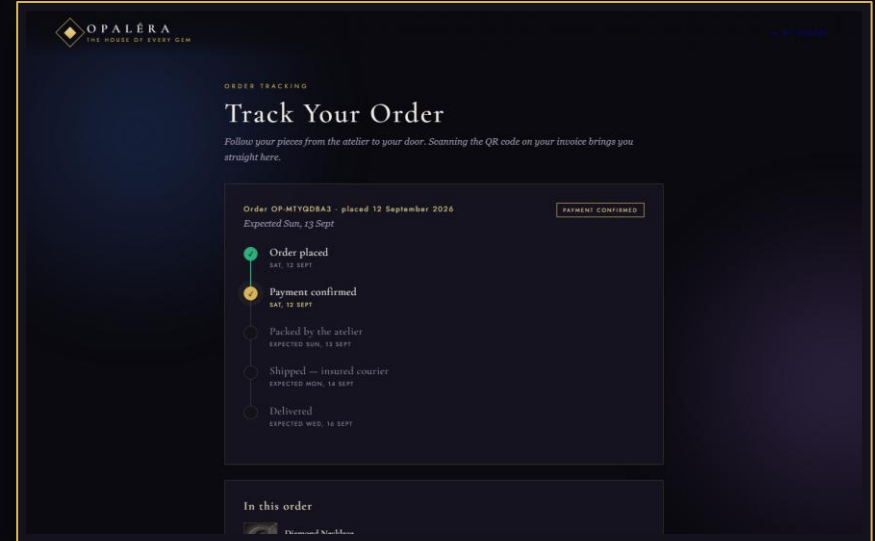
The screenshot displays the 'Payment' step of the checkout process for 'OPALÉRA THE HOUSE OF EVERY GEM'. The page is titled 'SECURE CHECKOUT' and 'Payment'. A progress bar shows four steps: 1. YOUR BAG, 2. DELIVERY & PAYMENT (active), 3. CONFIRMATION & INVOICE, and 4. A final step. The main content is divided into two columns. The left column, 'Your Order', lists two items: 'Heart Necklace' (R 1 399) and 'Sapphire Necklace' (R 3 518), each with a quantity selector set to 1 and 2 respectively. Below the items is a 'PROMO CODE (TRY: GEM10)' field with an 'APPLY' button. A checkbox for 'Wrap this order as a gift (complimentary)' is checked. A summary table shows: Items (R 4 917), VAT included (15%) (R 641), and Delivery (insured, tracked) (Free). The 'Total due' is R 4 917. The right column, 'Delivery Address', has fields for 'RECIPIENT FULL NAME' (filled with 'Demo Patron'), 'STREET ADDRESS' (filled with '12 Rosebank Avenue'), 'CITY / TOWN' (empty), 'POSTAL CODE' (empty), 'PROVINCE' (dropdown menu showing 'Gauteng'), and 'PHONE' (filled with '071 234 5678'). At the bottom right, a 'Payment' section shows 'for Demo Patron' with a card icon and a masked card number.

A Tax Invoice — and Where Your Order Is



Tax invoice · generated from the server's order record: letterhead, invoice №, date, method, line items with inscriptions, discount and VAT broken out, print / save as PDF.

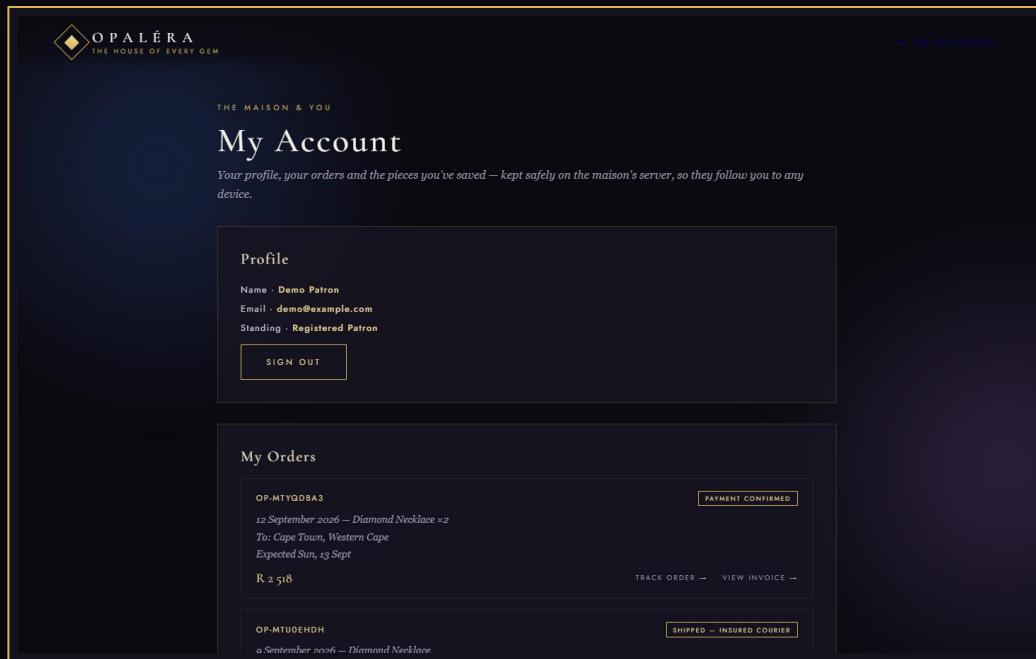
INVOICES · DONE



Tracking timeline · Placed → Paid → Packed → Shipped → Delivered (a collection path for pay-at-counter), advancing with time; the QR on the invoice opens it.

BONUS

Three Kinds of Visitor, Three Doors



- ◆ Guest — browses, searches, votes, reads reviews
- ◆ Patron — registers (bcrypt-hashed password, httpOnly cookie session); wishlist and bag saved on the server; orders, verified reviews, order history & invoices
- ◆ Manager — a separate console session (role “manager” on the server); patron accounts are refused
- ◆ Forgot-password: one-time 6-digit code, single use, 15-minute expiry; show-password eye on every field

REGISTRATION · DONE

LOGIN · DONE

ROLES · DONE

One Server, Eight Collections, 22 Endpoints



users

email · names · script salt+hash · role
· createdAt



sessions

token → email · expiry (patron &
manager scopes)



products

80 pieces: category · name · price ·
img · desc · stock · hidden



orders

ref · items · totals · VAT · promo · gift ·
address · method · status



reviews

pid · stars · text · verified · patron



wishlists · carts

per patron, merged on sign-in



resets

one-time codes with expiry & attempt
limit

*Reset the demo: delete db.json — the
catalogue re-seeds and the manager
account is recreated on the next run.*

AUTH

POST signup · login · forgot · reset · logout · GET me · admin/me

CATALOGUE

GET products · POST / PUT /:id / DELETE /:id (manager)

PATRON

GET/PUT cart · GET/PUT wishlist · GET/POST orders · GET reviews, reviews/:pid · POST reviews

MANAGER

GET admin/orders · GET admin/users

A Back Office in the Same World



- ◆ Own manager sign-in — verified by the server; patrons are refused
- ◆ Six KPI cards that count up: pieces sold with an improving / declining verdict, revenue, orders & average order, different products sold, stock on hand, new patrons
- ◆ Charts grow into place: units per month with the revenue line, sales-by-collection donut, units by province, users per day
- ◆ Live indicator, greeting and toasts — the same midnight-and-gold as the storefront

PRODUCT MGMT · DONE

REPORTS · DONE

Reports: Data Decides

- ◆ Filter by period (all time · 30 / 90 days · this year · custom months), collection, province and source — every figure recalculates
- ◆ Values: pieces sold + growth verdict, revenue, orders & average order, different products sold (text control), stock on hand & vault value by collection, registered users per day, best sellers
- ◆ Sample history is clearly labelled; live orders from the database merge on top
- ◆ Orders and Patrons tabs list every order and every account with sign-up dates



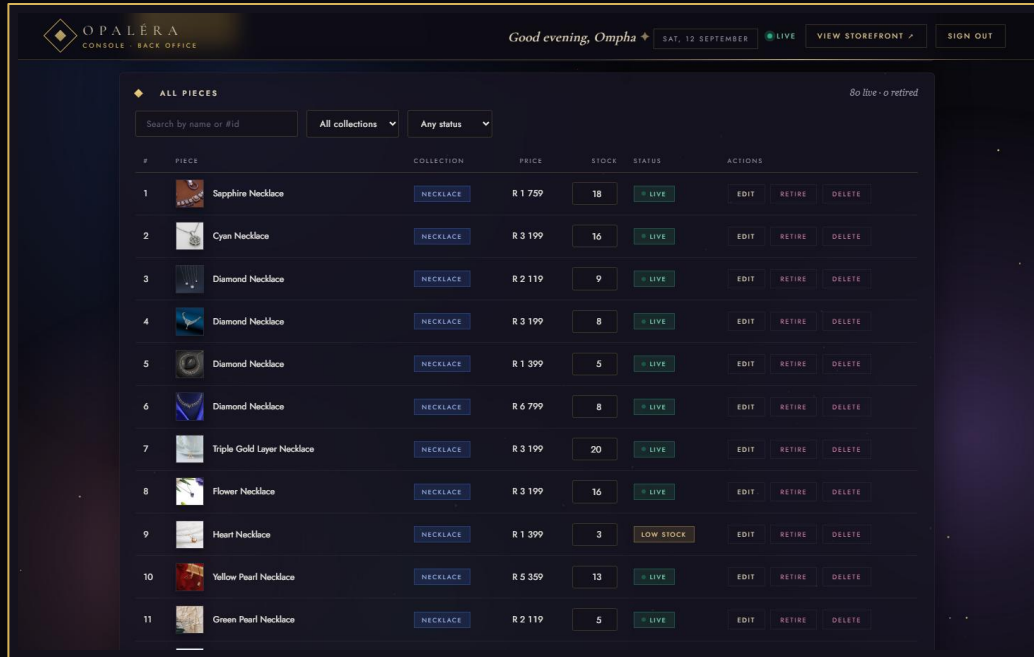
FILTERING

STOCK ON HAND

USERS / DAY

DISTINCT PRODUCTS

Add, Edit, Retire, Delete



Opaléra CONSOLE BACK OFFICE

Good evening, Ompha + SAT, 12 SEPTEMBER LIVE VIEW STOREFRONT SIGN OUT

◆ ALL PIECES *80 live · 0 retired*

Search by name or #id All collections Any status

#	PIECE	COLLECTION	PRICE	STOCK	STATUS	ACTIONS
1	 Sapphire Necklace	NECKLACE	R 1 759	18	LIVE	EDIT RETIRE DELETE
2	 Cyan Necklace	NECKLACE	R 3 199	16	LIVE	EDIT RETIRE DELETE
3	 Diamond Necklace	NECKLACE	R 2 119	9	LIVE	EDIT RETIRE DELETE
4	 Diamond Necklace	NECKLACE	R 3 199	8	LIVE	EDIT RETIRE DELETE
5	 Diamond Necklace	NECKLACE	R 1 399	5	LIVE	EDIT RETIRE DELETE
6	 Diamond Necklace	NECKLACE	R 6 799	8	LIVE	EDIT RETIRE DELETE
7	 Triple Gold Layer Necklace	NECKLACE	R 3 199	20	LIVE	EDIT RETIRE DELETE
8	 Flower Necklace	NECKLACE	R 3 199	16	LIVE	EDIT RETIRE DELETE
9	 Heart Necklace	NECKLACE	R 1 399	3	LOW STOCK	EDIT RETIRE DELETE
10	 Yellow Pearl Necklace	NECKLACE	R 5 359	13	LIVE	EDIT RETIRE DELETE
11	 Green Pearl Necklace	NECKLACE	R 2 119	5	LIVE	EDIT RETIRE DELETE

- ◆ Add a piece: name, collection, price, stock, photo (with preview) and description → POST /api/products
- ◆ Edit any piece in place — the storefront and its product page update immediately
- ◆ Inline stock control; low-stock and out-of-stock flags; retire hides a piece while keeping its history
- ◆ Delete removes it from the database; search and status filters over all 80 pieces

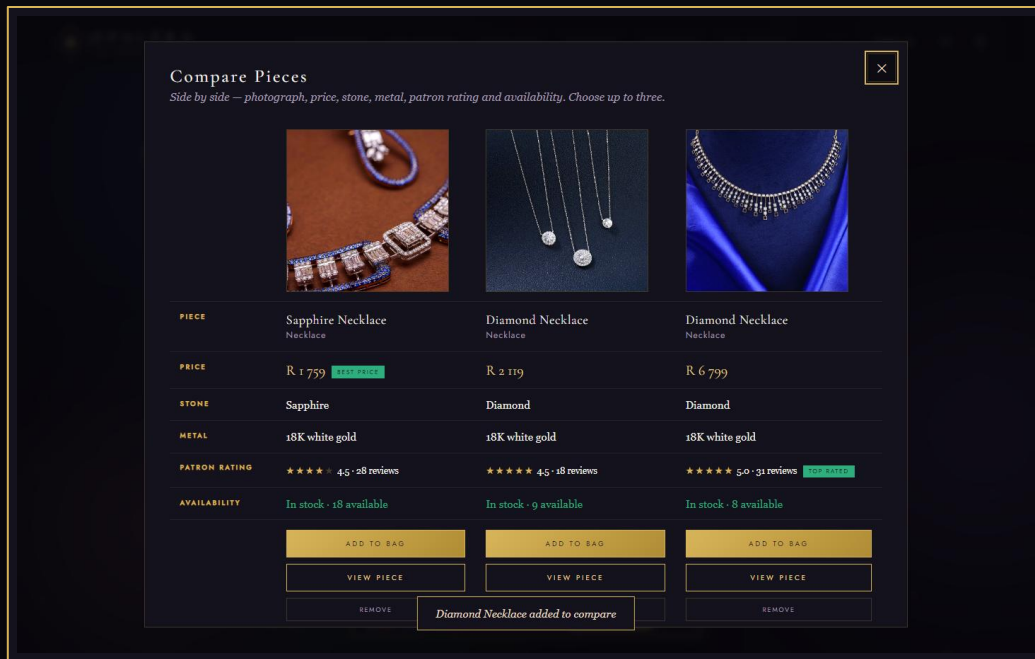
ADD · 3

EDIT · 5

DELETE · 2

Compare, Track, Scan

- ◆ Compare pieces — tick up to three; a side-by-side table of photo, price (“Best price”), stone, metal, rating (“Top rated”) and availability, with add-to-bag in place
- ◆ Order tracking — a live timeline per order, with status chips and ETA on the account page
- ◆ QR code on every invoice — printed invoices open the order's tracking page from a phone
- ◆ Also: the Studio customiser with content-filtered inscriptions, and a motion system that respects reduced-motion



Built to Be Trusted



54 automated API checks

Site & static blocking, sign-up / login / reset, cart, wishlist, orders with server-side totals and stock deduction, reviews, roles, product CRUD and validation — all passing.



Browser-verified pages

Every page loaded with zero console errors; compare, tracking, invoice QR, account and the console exercised end to end.



Security by default

script password hashes with per-user salt, constant-time comparison, httpOnly SameSite cookies, server-recomputed totals, the backend folder never served over HTTP, role checks on the server.



Accessible & considerate

Reduced-motion switch, ARIA tabs and live regions, focus rings, colour contrast checked; a multi-agent review found and fixed 29 defects before release.

Live on Render, Runs Anywhere Node Does

RUN IT LOCALLY

```
node opalera-backend/server.js
```

- ◆ Then open <http://localhost:3000> — it redirects to the storefront
- ◆ One server hosts the whole site and its API; no installs, no build step
- ◆ Data lives in `opalera-backend/data/db.json`, created on first run

DEMO ACCOUNTS

Patron demo@example.com / `demo123`

Manager

manager@opalera.co.za / `Opalera2026`



Live site

opalera.onrender.com

Storefront `/pages/jewellery_Html.html`

Console `/pages/admin.html`



Source

github.com/Thrivi17/Opalera

Render redeploys from the repository on every push.

ALL PIECES 80

NECKLACES 30

EARRINGS 20

PENDANTS 30

WISHLIST 0

THE STUDIO +

Search the archive

Every Price

Curated Order

SHOWING 80 OF 80 PIECES

OPALÉRA

THE HOUSE OF EVERY GEM

Sapphire Necklace

★★★★ 4.4 23

R 1 759

VIEW PIECE —

Cyan Necklace

★★★★ 4.5 28

Quantum Coders · Tsongorera RJN · Thedzisani KOP · Madhavan T · Mtshare B

Diamond Necklace

★★★★ 4.5 18

Diamond Necklace

★★★★ 4.3 20

R 3 199

VIEW PIECE —